

# SPONGES

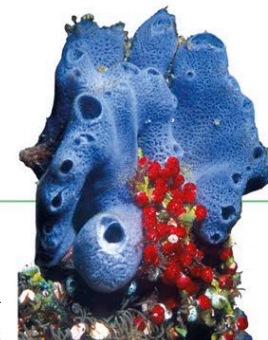
Simple, mostly marine animals, adult sponges live permanently attached to rocks, corals, and shipwrecks. A few species live in fresh water.

The Porifera vary in size and shape, from thin sheets to huge barrels, but all have the same basic structure: different types of specialist cells, but no organs. A system of water canals branches through the sponge, and water is drawn into them through

tiny pores in the sponge's surface. Special collar cells lining the water canals trap and engulf bacteria and other small plankton for food, and waste water exits through large openings called oscula.

Many sponges are classically "spongy", others are rock hard, soft, or even slimy depending on the supporting skeleton, which is made up of tiny spicules of silica or calcium carbonate. Spicules vary in shape and number between species and are an important feature in identification.

|          |                 |
|----------|-----------------|
| PHYLUM   | PORIFERA        |
| CLASSES  | 4               |
| ORDERS   | 32              |
| FAMILIES | 144             |
| SPECIES  | More than 9,000 |



**BLUE SPONGE**  
*Haliclona* sp.  
F: Chalinidae

One of very few blue-coloured sponges, this species commonly grows on the tops of corals and rocks off N. Borneo.

## CALCAREOUS SPONGES

The skeleton of Calcarea is composed of densely packed, mostly star-like, calcium carbonate spicules, each with three or four pointed rays. Variable in shape and crunchy to the touch, most species are small and lobe- or tube-shaped.



8 cm  
3 1/4 in

**LEMON SPONGE**  
*Leucetta chagosensis*  
F: Leucettidae

Growing on steep coral reefs in the W. Pacific, this sac-like sponge provides a splash of bright colour.



1–4 cm  
3/8–1 1/2 in

**CLATHRINA CLATHRUS**  
F: Clathrinidae

Made up of many tubes, each only a few millimetres wide, this Mediterranean sponge is a distinctive yellow colour.



2–5 cm  
1 3/4–2 in

**PURSE SPONGE**  
*Sycon ciliatum*  
F: Syctettidae

This simple, hollow sponge of N.E. Atlantic coasts has spiky calcareous spicules surrounding its osculum.

spicules  
surrounding  
osculum



10 cm  
4 in

tiny pores allow  
water entry

**LEUCONIA SP.**  
F: Baeriidae

The shape of this N.E. Atlantic sponge varies from lobes and cushions to crusts. It grows in areas of high water movement.



8 cm  
3 1/4 in

**RED PURSE SPONGE**  
*Grantessa* sp.  
F: Heteropiidae

Shaped like a small gourd, this delicate sponge grows between coral heads on shallow reefs in Malaysia and Indonesia.

## DEMOSPONGES

More than eighty-five per cent of all sponges belong to the Demospongiae. Although they are varied in appearance, most have a skeleton made up of scattered silicon dioxide spicules and a flexible organic collagen called spongin. A few encrusting species have no skeleton and some have only spongin.

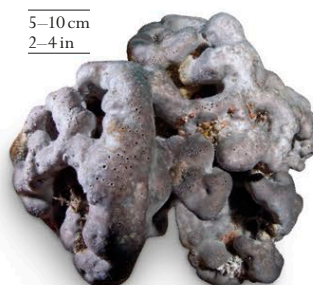


1 m  
3 1/4 ft

**BROWN TUBE SPONGE**  
*Agelas tubulata*  
F: Agelasidae

This sponge consists of uneven brown tubes arranged in clusters. It is common on deep reefs in the Caribbean and Bahamas.

5–10 cm  
2–4 in



**ELEPHANT HIDE SPONGE**  
*Pachymatisma johnstonia*  
F: Geodiidae

Mounds of this tough sponge can cover large areas of rocks and wrecks in clear coastal waters of the N.E. Atlantic.

30–40 cm  
12–16 in

**RED TREE SPONGE**  
*Negombata magnifica*  
F: Podospongiidae

Attempts to farm this beautiful Red Sea sponge are meeting with some success. It contains chemicals of potential medical importance.



35 cm  
14 in

**MEDITERRANEAN BATH SPONGE**

*Spongia (Spongia) officinalis*  
F: Spongiidae

This sponge has an elastic skeleton and no harsh spicules so after it has been cleaned and dried, it is ideal for washing with.



12 cm  
4 3/4 in

**GOLF BALL SPONGE**  
*Paratetilla bacca*  
F: Tetillidae

One of many tropical ball-shaped sponges, this species grows on sheltered coral reefs in the W. Pacific.